



JAYPEE UNIVERSITY OF ENGINEERING AND TECHNOLOGY, GUNA

Department of computer science

Minor project presentation

On

Soil Based Crop Recommendation for Madhya Pradesh

Under the Guidance

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Project no.:MK4

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Presentation Overview

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Introduction

- Agriculture is the backbone of India's economy, engaging nearly 60% of the population.
- Soil type and fertility play a vital role in determining crop yield.
- Farmers often rely on experience and traditional practices rather than scientific analysis.
- This leads to wrong crop choices, low productivity, and soil degradation.
- Our project introduces a Machine Learning-based system that uses soil and climatic data to recommend suitable crops, ensuring better yield and sustainable farming.

Problem Statement

- Farmers rely on traditional practices and guesswork.
- Wrong crop selection leads to poor yield & soil depletion.
- Existing advisory systems are limited in accuracy & coverage.
- Need for a system that uses soil data + ML for crop recommendation

Our solution

- A data-driven Crop Recommendation System that predicts suitable crops for given soil conditions. Use Soil + Environmental data (N, P, K, pH, rainfall, temperature, humidity).
- Apply Machine Learning model for accurate prediction.
- Reduces dependency on guesswork and improves yield accuracy.
- Integrates soil data analysis with machine learning prediction for precision agriculture.
- Can be further customized for region-specific or seasonal recommendations.
- Recommend the best suitable crop for given soil conditions.
- User-friendly system accessible to farmers.

Methodology

Workflow Steps:

1. Data Collection: Dataset collected from Kaggle and government agricultural sources.
2. Data Preprocessing: Missing values handled, features normalized, dataset balanced.
3. Model Training: Multiple models tested; Random Forest Classifier achieved best results.
4. Testing Phase: Model tested with unseen soil and weather data.
5. Prediction: Model predicts the most suitable crop based on input parameters.

Note: Random Forest combines multiple decision trees to improve accuracy and prevent overfitting.

Dataset

- Source: Kaggle – “Crop Recommendation Dataset.”
- Attributes (Inputs):
 - Nitrogen (N), Phosphorus (P), Potassium (K)
 - Temperature, Humidity, pH, Rainfall
- Target (Output): Crop Name (e.g., Rice, Maize, Cotton, etc.)
- Dataset Size: ~900 records.
- Dataset is balanced, ensuring fair training for all crop classes.
- Balanced data enhances model accuracy and generalization capability.

Random Forest Model (Introduction)

➡ **Random Forest** is a supervised machine learning algorithm used for both **classification** and **regression**

➡ Works by creating **multiple decision trees** and combining their outputs →

- **Majority vote** for classification.
- **Average prediction** for regression.

➡ • **Ensemble learning method** → improves accuracy by reducing overfitting.

- **Robust to noise & missing data** → works well with real-world ride datasets.

System Architecture

System Flow:

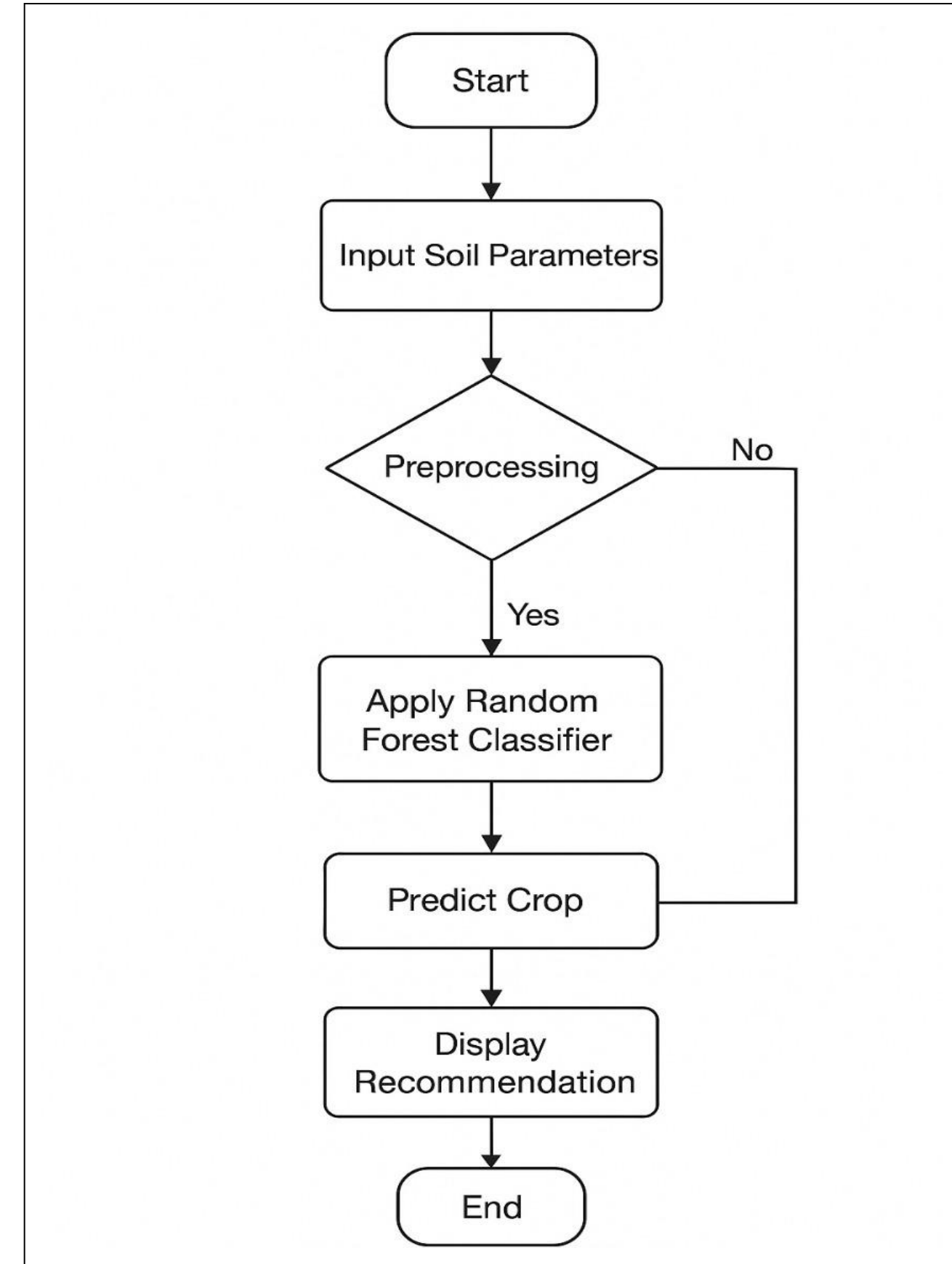
- Input: User enters soil and environmental parameters.
- Process: ML model analyzes data and compares with trained dataset.
- Output: System displays the recommended crop.

Technology Stack:

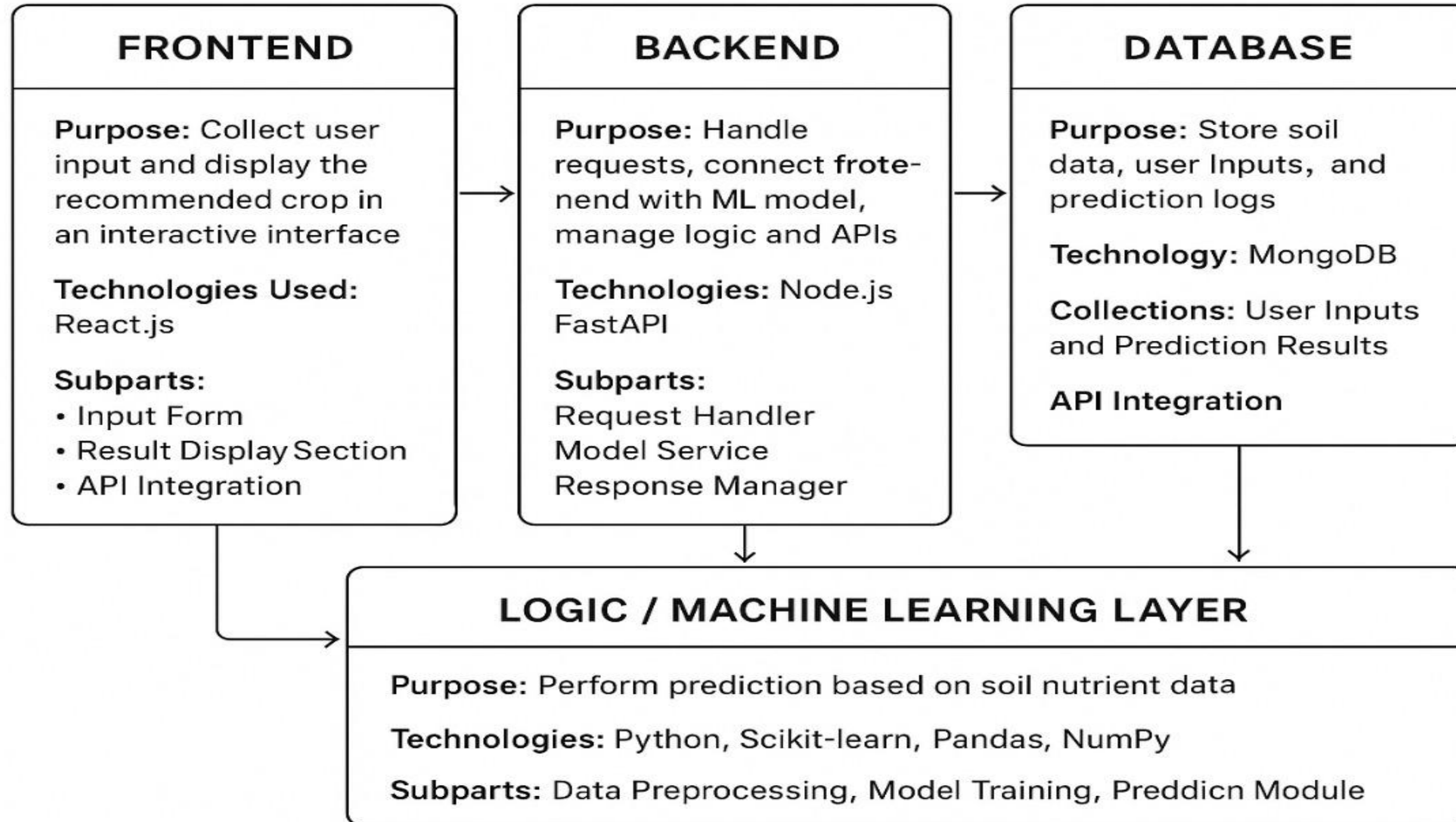
- Backend: Python (Flask / FastAPI).
- Libraries: scikit-learn (ML), pandas, numpy.
- Frontend: React, HTML, CSS.

Working:

Frontend collects inputs → sends to backend API → model processes data → returns crop recommendation.



Techstack



Evaluation Matrix

- To evaluate model performance, we used:
- Accuracy: Measures percentage of correct predictions.
- Precision & Recall: Evaluate reliability and completeness of predictions.
- F1-Score: Balances precision and recall for overall performance.
- Confusion Matrix: Visualizes true vs predicted results.
- Result:

Random Forest achieved 95% accuracy, indicating strong generalization and reliability for crop prediction.

Results, Conclusion & Future Work

Results:

- . Achieved 95% accuracy using Random Forest Classifier.
- . System effectively recommends suitable crops based on given inputs.

Conclusion:

- . Enables farmers to make data-backed and scientific decisions.
- . Improves productivity, reduces cost, and promotes sustainable farming by protecting soil health.

Future Work:

- . Integrate with real-time weather forecasting APIs.
- . Develop a mobile application with regional language support.

Thank you!